A First Course in Abstract Algebra

Section 51: Separable Extensions

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

51 Separable Extensions 3

Exercise 51.1: Combining square and cube roots 3

Exercise 51.2: Combining fourth and sixth roots 3

Exercise 51.3: A primitive biquadratic generator 3

Exercise 51.4: A primitive generator for a complex sextic field 3

Exercise 51.5: Where multiplicity is measured 4

Exercise 51.6: The definition of separable element 4

Exercise 51.7: A repeated polynomial over a perfect field 4

Exercise 51.8: Ten statements about separability 4

Exercise 51.9: Separable elements are closed under field operations 4

Exercise 51.10: A basis for a Frobenius extension 5

Exercise 51.11: Algebraic extensions of a perfect field 5

Exercise 51.12: Transitivity for infinite separable extensions 5

Exercise 51.13: The separable closure inside an algebraic extension 5

Exercise 51.14: Frobenius generates a finite-field Galois group 5

Exercise 51.15: The additive derivative homomorphism 5

Exercise 51.16: Formal derivative rules 6
Exercise 51.17: Multiplicity and vanishing derivative 6
Exercise 51.18: Characteristic zero forces irreducible separability 6
Exercise 51.19: The derivative test for an inseparable irreducible 6
Exercise 51.20: Common factors over the algebraic closure 6
Exercise 51.21: Bringing the gcd test back to the coefficient field 7
Exercise 51.22: A root-free computational test 7

51 Separable Extensions

Exercise 51.1: Combining square and cube roots

Choose $\alpha = \sqrt{2}/\sqrt[3]{2} = 2^{1/6}$. This quotient is in the given field. Direct exponent calculations give $\alpha^3 = \sqrt{2}$ and $\alpha^2 = \sqrt[3]{2}$, which are polynomials in α with rational coefficients. Thus each original generator lies in $\mathbb{Q}(\alpha)$, while α lies in their field, proving equality.

Exercise 51.2: Combining fourth and sixth roots

Choose $\alpha = \sqrt[4]{2}/\sqrt[6]{2} = 2^{1/12}$, an element of the given field. Then $\sqrt[4]{2} = \alpha^3$ and $\sqrt[6]{2} = \alpha^2$. These explicit polynomial expressions prove $\mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2}) = \mathbb{Q}(\alpha)$.

Exercise 51.3: A primitive biquadratic generator

Let $\alpha = \sqrt{2} + \sqrt{3}$, which belongs to the given field. Cubing and simplifying gives $\alpha^3 = 11\sqrt{2} + 9\sqrt{3}$. Subtracting suitable multiples of α yields

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{2}, \qquad \sqrt{3} = \frac{11\alpha - \alpha^3}{2}.$$

Thus both generators are rational polynomials in α, giving the desired equality.

Exercise 51.4: A primitive generator for a complex sextic field

Put $r = \sqrt[3]{2}$ and choose $\alpha = r + i$. From $(\alpha - i)^3 = 2$ and $i^2 = -1$,

$$i = \frac{\alpha^3 - 3\alpha - 2}{3\alpha^2 - 1}.$$

The denominator is nonzero: α has imaginary part 1 and positive real part, so its square has nonzero imaginary part. This already proves that $i, r \in \mathbb{Q}(\alpha)$. To meet the requested *polynomial* form, exact reduction gives

$$i = \frac{-12\alpha^5 + 9\alpha^4 - 40\alpha^3 + 78\alpha^2 - 78\alpha - 91}{22},$$
$$r = \frac{12\alpha^5 - 9\alpha^4 + 40\alpha^3 - 78\alpha^2 + 100\alpha + 91}{22}.$$

Here is a way to check these expressions without assuming the answer. Squaring the first displayed numerator equation for i gives

$$m(\alpha) = 0, \quad m(x) = x^6 + 3x^4 - 4x^3 + 3x^2 + 12x + 5.$$

Polynomial division verifies that $(3x^2 - 1)$ times the displayed polynomial for i is congruent to $x^3 - 3x - 2$ modulo $m(x)$. Evaluation at α proves the formula; subtracting it from α gives the formula for r. Hence the field is $\mathbb{Q}(\alpha)$.

Exercise 51.5: Where multiplicity is measured

The factorization must be in $\overline{F}[x]$, not necessarily in $F[x]$, since α may lie outside F. For a *nonzero* polynomial f, multiplicity $v \geq 1$ means $(x-\alpha)^v \mid f$ but $(x-\alpha)^{v+1} \nmid f$ in $\overline{F}[x]$. The zero polynomial has no finite root multiplicity under this definition.

Exercise 51.6: The definition of separable element

The proposed definition is correct: an algebraic element is separable over F when it has multiplicity 1 as a root of its minimal polynomial over F. All roots of an irreducible polynomial have the same multiplicity, so this also says the minimal polynomial has distinct roots in an algebraic closure.

Exercise 51.7: A repeated polynomial over a perfect field

Take $f(x) = (x^2+1)^2$. It has no rational root, and its complex roots $i, -i$ each have multiplicity 2. This does not contradict perfection of $\mathbb{Q}$, which concerns *irreducible* polynomials. The minimal polynomial x^2+1 has distinct roots; the repetitions were introduced by squaring it.

Exercise 51.8: Ten statements about separability

(a) False. Over $\mathbb{Z}_p(t)$, $x^p - t$ is irreducible and has just one distinct root, so adjoining that root gives an inseparable finite extension.

(b) True. Every finite field is perfect.

(c) True. An irreducible polynomial in characteristic zero cannot share a factor with its smaller-degree nonzero derivative.

(d) False. $(x-1)^2$ has a repeated root over any field.

(e) False for the same example, even over $\mathbb{Q}$.

(f) True. This is the defining polynomial characterization of a perfect field.

(g) True. Every irreducible polynomial over an algebraically closed field is linear.

(h) True. An algebraic closure is algebraic over F and is perfect.

(i) True. Separability makes the embedding index equal the degree, and splitting makes every embedding an automorphism.

(j) True. The index of a finite extension divides its degree (the quotient is its inseparable degree), and in a splitting field the index equals the group order.

Exercise 51.9: Separable elements are closed under field operations

The minimal polynomial of β over $F(\alpha)$ divides its separable minimal polynomial over F, so it is separable. Therefore $F(\alpha,\beta)/F(\alpha)$ and $F(\alpha)/F$ are finite separable extensions; the tower theorem makes $F(\alpha,\beta)/F$ separable. Every element of this field is then separable over F, including $\alpha+\beta$, $\alpha\beta$, and α/β when $\beta \neq 0$.

Exercise 51.10: A basis for a Frobenius extension

Let $K = \mathbb{Z}_p(y^p)$. The relation $y^p \in K$ makes $K[y]$ spanned by $1, y, \ldots, y^{p-1}$. Being a finite-dimensional domain over K, it is a field: multiplication by any nonzero element is an injective linear map, hence surjective, giving an inverse. Thus $K[y] = \mathbb{Z}_p(y)$. For independence, a relation $\sum_{j=0}^{p-1} r_j(y^p)y^j = 0$ can be multiplied by a common denominator to give $\sum_j P_j(y^p)y^j = 0$. Exponents coming from different j lie in different residue classes modulo p, so no cancellation between the groups is possible; all P_j vanish. Thus the displayed spanning set is a basis, and the degree is p. Setting $t = y^p$, the polynomial $x^p - t$ has y as a root and the same degree as its minimal polynomial. Hence it is irreducible over $\mathbb{Z}_p(t)$.

Exercise 51.11: Algebraic extensions of a perfect field

Let α be algebraic over E. Since E/F is algebraic, α is algebraic over F as well. The minimal polynomial of α over the perfect field F is separable. Its minimal polynomial over E divides it in $E[x]$, so also has no repeated roots. Thus every element algebraic over E is separable over E, proving E perfect.

Exercise 51.12: Transitivity for infinite separable extensions

Take $\alpha \in K$. Its separable minimal polynomial over E uses finitely many coefficients $c_1, \ldots, c_m \in E$. Set $E_0 = F(c_1, \ldots, c_m)$; this is a finite separable extension of F. The same polynomial is irreducible over E_0, because a factorization there would be one over E, and it remains separable. Thus $E_0(\alpha)/E_0$ is finite separable. The finite tower theorem gives $E_0(\alpha)/F$ separable, so α is separable over F. This holds for every $\alpha \in K$, proving transitivity without assuming the entire tower finite.

Exercise 51.13: The separable closure inside an algebraic extension

Let S be the elements of E separable over F. It contains F, including 0 and 1. Exercise 51.9 shows it is closed under sums, products, and quotients by nonzero elements; additive inverses are included by multiplying by $-1 \in F$. Hence S is a subfield. It consists exactly of the separable elements, so is the largest intermediate extension separable over F, called the separable closure of F in E.

Exercise 51.14: Frobenius generates a finite-field Galois group

(a) In a field of p^n elements, $a^{p^n} = a$ for every a, so $\sigma_p^n = 1$. If $\sigma_p^d = 1$ for $0 < d < n$, all p^n elements would be roots of $x^{p^d} - x$, whose degree p^d is smaller than p^n, impossible. Thus its order is n.

(b) The extension has degree n, is separable because finite fields are perfect, and is a splitting extension since it splits $x^{p^n} - x$. Its Galois group therefore has order n. The n distinct powers of Frobenius already account for that many elements, so the full group is cyclic with this generator.

Exercise 51.15: The additive derivative homomorphism

(a) If $f = \sum a_j x^j$ and $g = \sum b_j x^j$, differentiating coefficient by coefficient gives $(f+g)' = \sum j(a_j + b_j)x^{j-1} = f' + g'$. This proves it is a homomorphism of additive groups; it is generally not a

ring homomorphism.

(b) In characteristic zero, $ja_j = 0$ for $j > 0$ implies $a_j = 0$. Thus the kernel is the constant polynomials F.

(c) In characteristic p, j vanishes as a field scalar precisely when $p \mid j$. Hence the kernel is $F[x^p]$, the polynomials all of whose nonzero exponents are multiples of p. Their coefficients need not themselves be pth powers.

Exercise 51.16: Formal derivative rules

(a) For $c \in F$, coefficientwise differentiation gives $(cf)' = \sum_j jca_j x^{j-1} = cf'$.

(b) First check $(x^i x^j)' = (i+j)x^{i+j-1} = x^i(jx^{j-1}) + (ix^{i-1})x^j$. Expand arbitrary polynomials into monomials and use additivity and (a) to obtain $(fg)' = fg' + f'g$.

(c) Induct on m. For $m = 1$ the assertion is immediate. If $(f^m)' = mf^{m-1}f'$, the product rule gives $(f^{m+1})' = f^m f' + (f^m)'f = (m+1)f^m f'$. The integers are interpreted as elements of F, so this proof also works in positive characteristic.

Exercise 51.17: Multiplicity and vanishing derivative

In $\overline{F}[x]$, write $f = (x-\alpha)^v g$ with $g(\alpha) \neq 0$. If $v = 1$, the product rule gives $f'(\alpha) = g(\alpha) \neq 0$. If $v > 1$, both terms of

$$f' = v(x-\alpha)^{v-1}g + (x-\alpha)^v g'$$

vanish at α, including when the scalar v is zero in the field. Thus $f'(\alpha) = 0$ exactly when the root has multiplicity greater than 1.

Exercise 51.18: Characteristic zero forces irreducible separability

Let $p \in F[x]$ be nonconstant and irreducible in characteristic zero. Its derivative is nonzero and has smaller degree. If a root α were repeated, it would be a root of p' too, so its minimal polynomial p would divide p'. The degree inequality rules this out. Therefore all roots of p are simple, proving separability.

Exercise 51.19: The derivative test for an inseparable irreducible

For an irreducible q, a repeated root implies $q \mid q'$ by minimality. Since $\deg q' < \deg q$ unless $q' = 0$, this is equivalent to $q' = 0$. In characteristic p, coefficientwise differentiation gives $q' = 0$ exactly when each exponent with nonzero coefficient is divisible by p. Conversely a nonconstant polynomial with zero derivative cannot have a simple root by Exercise 51.17. Thus the criterion holds in both directions.

Exercise 51.20: Common factors over the algebraic closure

By Exercise 51.17 a repeated root is exactly a common root of f and f'. Over the algebraically closed field $\overline{F}$, a common root α supplies the common linear factor $x - \alpha$. Conversely a nonconstant

common factor has a root in the closure, which is a common root of f, f', hence repeated in f. This proves the requested equivalence for nonzero f.

Exercise 51.21: Bringing the gcd test back to the coefficient field

If f, f' have no nonconstant common divisor in $F[x]$, their gcd is 1, so Bezout's identity gives $Af + Bf' = 1$ with $A, B \in F[x]$. This same identity holds in $\overline{F}[x]$, excluding any nonconstant common factor there. Conversely a nonconstant common factor in $F[x]$ remains one over the closure. Combining with Exercise 51.20 proves the equivalence over F itself. The argument includes nonzero constant f, whose gcd with its zero derivative is a unit.

Exercise 51.22: A root-free computational test

For a nonzero polynomial, form its formal derivative and run the Euclidean algorithm in $F[x]$ on f, f'. Normalize the final nonzero remainder to a monic gcd d. A repeated root exists exactly when $\deg d > 0$. In particular, if f is nonconstant and $f' = 0$, the gcd is the monic associate of f, so there is repetition. If f is a nonzero constant, the gcd has degree 0 and there are no roots. No numerical root approximation is required.